

HighPhytoSparseNet: A Lightweight Multi-Head Object Detection Model for Edge-Based Agricultural Applications

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Abstract—Object detection in agricultural environments presents unique challenges, including scale variation, class imbalance, and limited annotated data. Although modern deep learning-based detectors achieve high performance on large-scale benchmarks, their computational complexity often restricts deployment on edge devices. In this work, we propose PhytoNetEdge, a lightweight anchor-based single-stage object detection framework designed for real-time botanical applications. The proposed architecture employs a MobileNetV3-Small backbone for efficient feature extraction, followed by a bidirectional feature pyramid network to enhance multi-scale representation. A three-head detection module is introduced to specialize in small, medium, and large object detection, with a head-weighted loss strategy to improve sensitivity toward fine plant structures. The model contains approximately 3.8 million parameters and requires only 0.5 GFLOPs, making it suitable for edge deployment. Experimental results demonstrate that PhytoNetEdge achieves competitive detection accuracy while maintaining low computational overhead, highlighting its effectiveness for real-time agricultural monitoring systems.

Index Terms—Object detection, lightweight neural networks, edge computing, agricultural monitoring, multi-scale detection, MobileNetV3

I. INTRODUCTION

Recent advancements in artificial intelligence have significantly accelerated the development of object detection methods within computer vision. In particular, agricultural monitoring has emerged as a critical application domain, where automated visual analysis enables crop health assessment, yield estimation, and disease detection. Despite substantial progress in detection accuracy, deploying such models efficiently on resource-constrained edge devices remains a significant challenge due to computational and energy limitations.

Saqib et al. [1] provide a comprehensive review of deep learning-based object detection techniques for multimedia and agricultural applications, highlighting the evolution of two-stage detectors derived from R-CNN and their region proposal mechanisms such as Edge Boxes. In contrast, single-

stage detectors such as YOLO (You Only Look Once) [2] perform localization and classification in a unified forward pass, enabling faster inference suitable for real-time applications. Chetan et al. [3] further explore YOLO variants for agricultural object detection, demonstrating their effectiveness and structural efficiency for edge-oriented deployment.

More recent approaches incorporate transformer-based enhancements to improve detection robustness and feature representation. For example, Xue Xia et al. [4] propose MTY-OLOX, a multi-transformer-enabled architecture for tree-level apple inflorescence detection. Similarly, Helizani et al. [5] employ YOLOv3-tiny for coffee fruit detection during harvest, emphasizing lightweight design considerations.

Although modern object detection frameworks are well-developed and adaptable to various agricultural scenarios, a fundamental trade-off persists between detection accuracy and computational efficiency. Many high-performance models incur substantial parameter counts and inference costs, limiting their suitability for real-time deployment on edge hardware. Consequently, achieving an optimal balance between accuracy, model complexity, and inference speed remains an open research challenge.

In this work, we build upon established detection architectures and introduce HighPhytoSparseNet, a lightweight object detection framework designed for efficient agricultural edge deployment. The proposed model is optimized for real-time detection of tomato plants, specifically targeting stem and fruit localization under field-like conditions.

A custom agricultural dataset was constructed comprising over 2500 training images, along with 100 validation and 100 test samples. The dataset includes annotated instances of tomato fruits and stems captured under varying environmental conditions. The model is currently being trained and evaluated to assess detection accuracy, inference efficiency, and suitability for edge hardware deployment.

II. RELATED WORK

Object detection frameworks are broadly categorized into two-stage and single-stage architectures, each offering different trade-offs between detection accuracy and computational efficiency. Two-stage detectors first generate region proposals and subsequently perform classification and bounding box regression. While these approaches achieve high detection accuracy, their multi-step processing pipeline introduces substantial computational overhead.

A. Two-Stage Object Detectors

Two-stage object detection frameworks achieve high localization accuracy by decomposing detection into region proposal generation followed by classification and bounding box regression. The seminal R-CNN introduced deep convolutional neural networks for object detection by applying selective search to generate candidate regions and subsequently classifying each proposal independently [6]. Although this approach significantly improved detection accuracy over traditional methods, repeated convolutional processing for each region resulted in substantial computational redundancy. To mitigate this limitation, SPP-Net enabled feature sharing through spatial pyramid pooling, reducing repeated computation. This was further refined by Fast R-CNN [7], which unified training and introduced multi-task loss optimization for joint classification and localization. The introduction of the Region Proposal Network in Faster R-CNN enabled end-to-end training and improved proposal efficiency, while R-FCN reformulated the detection pipeline into a fully convolutional architecture using position-sensitive score maps to reduce per-region computation. Despite these architectural optimizations, the sequential nature of proposal generation followed by refinement continues to impose computational and latency constraints, limiting suitability for real-time edge deployment scenarios [8].

B. Single-Stage Object Detectors

In contrast, single-stage object detectors eliminate the explicit region proposal stage and perform classification and localization within a unified forward pass. One of the most influential single-stage frameworks is YOLO, which formulates object detection as a regression problem over grid-based predictions. The original YOLO architecture employed a deep convolutional backbone followed by fully connected layers to directly predict bounding box coordinates and class probabilities in a single evaluation [9]. While it achieved real-time inference speed, early versions exhibited localization errors and relatively weaker performance on small objects due to coarse grid partitioning. Subsequent improvements in the YOLO family enhanced feature extraction, multi-scale detection, and computational efficiency, with later variants such as YOLOv5 emphasizing lightweight design and improved accuracy for practical deployment scenarios [10]. Another prominent single-stage framework, SSD [11], introduced multi-scale feature maps to detect objects of varying sizes while maintaining high inference speed. By utilizing

default anchor boxes at different scales and aspect ratios, SSD improved small-object detection compared to earlier YOLO models and demonstrated competitive performance on benchmark datasets such as PASCAL VOC and COCO. Further advancements such as RetinaNet [12] addressed the class imbalance problem inherent in dense prediction by introducing focal loss, enabling single-stage detectors to achieve accuracy comparable to two-stage approaches while preserving computational efficiency. Owing to their streamlined architecture and reduced latency, single-stage detectors have become particularly suitable for real-time and edge-based applications.

C. Lightweight and Edge-Oriented Architectures

Lightweight object detection models have gained significant attention in recent years, particularly for deployment in Internet of Things (IoT) and edge-based vision applications where computational resources, memory, and power consumption are constrained. Efficient backbone architectures play a central role in enabling such deployments. For instance, MobileNet [13] introduced depthwise separable convolutions to drastically reduce computational complexity compared to standard convolutional operations. By incorporating width and resolution multipliers, MobileNet enables a flexible trade-off between model size and accuracy, making it suitable for embedded and mobile environments.

Building upon this, MobileNetV3 further optimized lightweight network design through neural architecture search and hardware-aware optimization. By integrating squeeze-and-excitation modules and hard-swish activation functions, MobileNetV3 achieves an improved accuracy-latency balance, making it particularly effective for real-time edge applications.

Similarly, ShuffleNet [14] proposed an efficient architecture for mobile devices by employing group convolutions and channel shuffle operations. These mechanisms significantly reduce parameter count and computational overhead while maintaining strong representational capacity. Such architectural innovations demonstrate how efficient feature extraction can be achieved without substantial performance degradation.

Beyond backbone-level optimization, detection frameworks have also focused on efficiency-driven design. EfficientDet [15] introduced compound scaling and a bidirectional feature pyramid network (BiFPN) to systematically balance depth, width, and resolution, achieving state-of-the-art accuracy with fewer parameters and FLOPs. Furthermore, lightweight variants of YOLO have been proposed to reduce inference latency while preserving competitive detection performance. These developments collectively emphasize the importance of designing computationally efficient detection systems suitable for deployment in resource-constrained edge environments.

Recent studies have also explored the integration of Generative Adversarial Networks (GANs) to enhance object detection performance under challenging conditions such as limited data availability, domain shift, and low illumination. The original

TABLE I: HighPhytoSparseNet Architectural Components

Component	Description
Backbone	MobileNetV3-Small (pretrained)
Neck	Bi-directional FPN (128 channels)
Detection Heads	3 (Small, Medium, Large)
Detection Type	Anchor-based, Single-stage

Generative Adversarial Network framework introduced adversarial training for realistic data generation, which has since been adopted for data augmentation and feature refinement in detection pipelines. While GAN-based approaches can improve robustness and generalization, they often introduce additional training complexity and computational overhead, making them less suitable for lightweight real-time edge deployment scenarios [16].

Despite significant advancements in computational power and deep learning techniques for computer vision and object detection, there remains considerable scope for improvement in domain-specific applications. Most existing models achieve high performance on large-scale benchmark datasets with abundant annotated samples; however, detection accuracy often degrades when dealing with rare classes, limited training data, or small and subtle objects. This issue is particularly prominent in agricultural environments, where fine-grained visual differences and scale variations pose additional challenges. To address these limitations, we propose HighPhytoSparseNet, a lightweight convolutional neural network-based object detection framework tailored for plant-based applications. The proposed architecture incorporates a multi-head detection and classification strategy to enhance feature representation while maintaining computational efficiency. The detailed architecture and design methodology are presented in the following section

III. PROPOSED METHODOLOGY

A. HighPhytoSparseNet Architecture

1) *Architecture Overview*: HighPhytoSparseNet is a lightweight multi-scale object detection framework designed for real-time botanical object detection on edge devices. The architecture follows a single-stage, anchor-based detection paradigm inspired by YOLO-based detectors [17]. The network is optimized to improve detection performance on small and medium-scale agricultural objects under dataset imbalance conditions.

The model consists of three major components: a lightweight backbone, a bidirectional feature aggregation neck, and three scale-specific detection heads. The overall architectural composition is summarized in Table I.

The design emphasizes feature reuse, multi-scale representation, and class imbalance mitigation, which are critical in agricultural detection scenarios where object size variation and skewed class distribution are common challenges. The anchor-based formulation was retained due to its stability in small-object localization under limited data regimes. Furthermore, transformer-based detectors were avoided due to higher com-

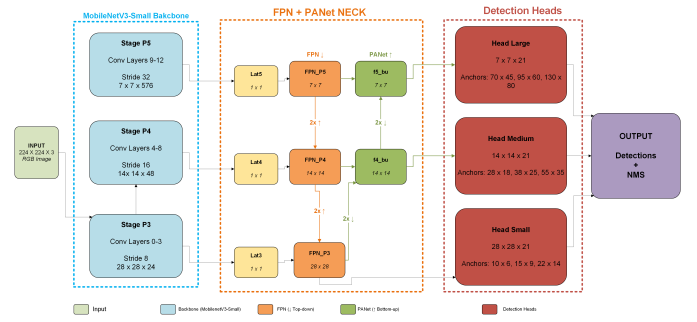


Fig. 1: Overall architecture of HighPhytoSparseNet. The model consists of a MobileNetV3-Small backbone, a bi-directional FPN-PANet neck, and three scale-specific detection heads.

TABLE II: Extracted Feature Map Resolutions

Level	Resolution	Stride	Purpose
P3	28×28	8	Small objects
P4	14×14	16	Medium objects
P5	7×7	32	Large objects

TABLE III: Bidirectional Feature Aggregation Design

Pathway	Function
Top-down FPN	Semantic enrichment
Bottom-up PANet	Localization refinement
Channels	128 per scale

putational overhead, which contradicts the edge-deployment objective.

2) *Backbone Network*: The backbone network is based on MobileNetV3-Small [?], pretrained on ImageNet. The network contains approximately 2.5M parameters and employs depthwise separable convolutions, squeeze-and-excitation (SE) modules [?], and hard-swish activation to reduce computational complexity while maintaining representational strength.

Multi-scale feature maps are extracted at three spatial resolutions, summarized in Table II.

These hierarchical representations enable robust multi-scale detection performance. MobileNetV3-Small was selected over larger backbones (e.g., ResNet-50) due to its favorable trade-off between parameter efficiency and representational capacity, making it suitable for edge-constrained environments.

3) *Feature Aggregation Neck*: To enhance cross-scale feature fusion, a bidirectional Feature Pyramid Network (FPN) [18] combined with a Path Aggregation Network (PANet) [19] is employed. The neck operates with 128-channel feature maps across all pyramid levels.

The bidirectional aggregation improves both semantic consistency (top-down) and localization precision (bottom-up), which is particularly beneficial for thin agricultural structures such as stems.

4) *Multi-Head Detection Module*: Three scale-specific detection heads are employed, each associated with three anchor boxes. The anchor configurations are dataset-adapted to capture tomato and stem objects efficiently.

TABLE IV: Detection Head Configuration

Head	Resolution	Optimized For	Anchors (pixels)
Small	28×28	Thin stems	(10×6), (15×9), (22×14)
Medium	14×14	Medium objects	(28×18), (38×25), (55×35)
Large	7×7	Large tomatoes	(70×45), (95×60), (130×80)

TABLE V: Loss Configuration

Component	Configuration
Localization Loss	CIoU [?]
Objectness Loss	BCE (cap = 50)
Classification Loss	Focal Loss ($\alpha = 0.25, \gamma = 2.0$) [?]
Class Weights	[4.0, 1.0]
λ_{box}	5.0
λ_{obj}	2.0
λ_{cls}	1.0
Head Weights (w_s, w_m, w_l)	(0.5, 0.35, 0.15)

TABLE VI: Training Configuration

Parameter	Value
Input Size	224×224
Epochs	500
Batch Size	16
Optimizer	AdamW (wd= 1×10^{-5})
Backbone LR	1×10^{-4}
Head LR	4×10^{-3}
Scheduler	OneCycleLR
Warmup	5%
Confidence Threshold	0.25
NMS IoU	0.45

TABLE VII: Data Augmentation Strategy

Technique	Purpose
Horizontal Flip	Orientation robustness
Vertical Flip	Viewpoint variation
RandomBrightnessContrast	Illumination robustness
HueSaturationValue	Color invariance
RandomScale	Scale robustness
Perspective	Geometric distortion tolerance

5) *Loss Function*: The total detection loss is formulated as a weighted combination of localization, objectness, and classification losses:

$$L_{total} = \sum_{h \in \{s, m, l\}} w_h (\lambda_{box} L_{box}^h + \lambda_{obj} L_{obj}^h + \lambda_{cls} L_{cls}^h) \quad (1)$$

where $h \in \{s, m, l\}$ denotes the detection heads.

The head weights were empirically selected to emphasize small-object detection, as stems represent the minority yet critical class in the dataset. Focal loss was adopted to mitigate class imbalance between tomato and stem categories.

6) *Training Strategy*: Differential learning rates were employed to preserve pretrained backbone representations while enabling faster adaptation of detection heads.

The training duration of 500 epochs was selected to ensure convergence under class imbalance conditions and extensive data augmentation.

7) *Data Augmentation*: Weighted random sampling with an 8× factor was applied to stem-containing images to mitigate dataset imbalance.

8) *Inference Procedure*:

TABLE VIII: Inference Configuration

Component	Setting
Multi-head Decoding	Per-head anchor assignment
Confidence Threshold	0.25
NMS IoU Threshold	0.45
NMS Type	Cross-head suppression

TABLE IX: Model Specifications of HighPhytoSparseNet

Metric	Value
Parameters	~3.8M
Model Size (FP32)	~15 MB
Model Size (INT8)	~4 MB
Input Size	224×224
GFLOPs	~0.5

9) *Model Specifications*: The compact parameter count and low computational complexity enable real-time deployment on edge devices.

IV. EXPERIMENTS AND RESULTS

In this section, we briefly discuss the experimental evaluation of our proposed HighPhytoSparseNet model. We first describe the dataset and implementation details, followed by evaluation metrics, quantitative results, comparative results with existing lightweight detectors, and ablation studies to validate architectural design choices.

A. Dataset Description

The proposed model was evaluated on a custom agricultural dataset, specifically curated for tomato plant monitoring under greenhouse conditions. We focus on two object categories: **stem** and **tomatoes**, representing structurally thin objects, and moderately sized fruit instances, respectively.

The dataset consists of annotated RGB images captured under varying illumination conditions, occlusions, and background complexities. Images are resized to a fixed resolution of 224×224 pixels during the training and evaluation. We use Bounding Box annotations based on the Yolo structure, enabling direct compatibility with anchor-based object detection frameworks.

The dataset is divided into training, validation and test subsets following a standard split protocol to ensure unbiased performance evaluation. Data augmentation techniques such as horizontal flipping, random scaling, and color jittering are applied to improve generalization and robustness against environmental variability.

A severe class imbalance and significant scale variation are observed in the dataset. Stem instances occupy substantially smaller spatial regions compared to tomatoes, making small-object detection particularly challenging. Moreover, stems are frequently occluded by foliage, increasing ambiguity and the likelihood of misclassification. These characteristics make the dataset a realistic benchmark for evaluating lightweight real-time object detection models.

The training split consists of 2,250 annotated images with a total of 24,850 object instances. A significant class imbalance

TABLE X: Tomato_d Dataset Overview (Training Split)

Class	Number of Instances	Percentage
Stem (Class 0)	4,294	17.3%
Tomato (Class 1)	20,556	82.7%
Total	24,850	100%



(a) Indoor tomato cluster



(b) Medium-scale fruit instances



(c) Outdoor stem and tomato cluster

Fig. 2: Representative samples from the tomato-stem dataset showing illumination variation, scale diversity, background clutter, and natural occlusion.

is observed, with tomato instances representing 82.7% of total annotations, motivating the use of class-weighted loss during training.

Stem instances are generally smaller and thinner compared to tomato objects. The median stem bounding box size (16.6×9.9 pixels) confirms the small-object detection challenge. Both classes exhibit near-square aspect ratios on average, though stems show greater variability due to elongated structures.

The percentile analysis highlights the prevalence of extremely small stem objects, with 10% of stems having widths below 8 pixels. This validates the need for high-resolution feature maps (28×28 detection head) for improved small-object sensitivity.

The anchor IoU analysis demonstrates that the original square anchors provide stronger matching for stems at IoU ≥ 0.3 , while the modified thin anchors maintain high coverage at lower IoU thresholds. This trade-off motivated the balanced anchor design adopted in HighPhytoSparseNet.

B. Implementation Details

The proposed HighPhytoSparseNet architecture, described in Section III, was implemented using PyTorch and trained on an NVIDIA GeForce RTX 4090 GPU. Automatic Mixed Precision (AMP) was enabled to accelerate computation and improve memory efficiency during training.

All experiments were conducted with an input resolution of 224×224 pixels. The model was trained for a maximum of 500 epochs with early stopping patience set to 250 epochs to mitigate overfitting. A batch size of 16 was employed with gradient accumulation of 2, resulting in an effective batch size of 32.

The MobileNetV3-Small backbone was initialized with ImageNet pretrained weights. To preserve learned feature representations, differential learning rates were applied: 1×10^{-4} for the backbone and 1×10^{-3} for the neck and detection heads.

Optimization was performed using the AdamW optimizer. The confidence threshold during training was set to 0.35, while evaluation was performed with a confidence threshold of 0.20 and an IoU threshold of 0.45.

Class imbalance between stem and tomato categories was addressed using weighted classification loss, and multi-head loss balancing was applied as described in Section III.

C. Evaluation Metrics

The performance of HighPhytoSparseNet was evaluated using standard object detection metrics, including Precision, Recall, F1-score, and mean Average Precision (mAP). In addition, computational efficiency was measured using inference speed (FPS).

1) Precision and Recall:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

where TP denotes true positives, FP false positives, and FN false negatives.

Precision measures the correctness of predicted detections, while Recall quantifies the model's ability to detect all ground-truth objects.

2) *F1-Score*: The F1-score is computed as the harmonic mean of Precision and Recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

This metric provides a balanced measure when dealing with class imbalance.

3) *Mean Average Precision (mAP)*: Average Precision (AP) is defined as the area under the Precision-Recall (PR) curve:

$$AP = \int_0^1 P(R) dR \quad (5)$$

The mean Average Precision is defined as:

TABLE XI: Bounding Box Size Distribution (in pixels, 224×224 resolution)

Class	Mean W	Mean H	Median W	Median H	Mean W/H	Max W
Stem	20.8	11.1	16.6	9.9	1.89	110.6
Tomato	27.9	16.5	24.0	14.5	1.74	224.0

TABLE XII: Bounding Box Percentiles (Width × Height in pixels)

Class	P10	P50 (Median)	P90
Stem	7.9 × 5.0	16.6 × 9.9	40.7 × 19.7
Tomato	15.0 × 8.6	24.0 × 14.5	33.5 × 20.4

TABLE XIII: Anchor Matching Quality (Mean Best IoU)

Anchor Set	Class	Mean IoU	% IoU ≥ 0.3	% IoU ≥ 0.15
Old Anchors	Stem	0.4933	90.1%	95.3%
Old Anchors	Tomato	0.5587	99.4%	99.9%
New Anchors	Stem	0.4063	66.5%	95.1%
New Anchors	Tomato	0.5528	94.2%	99.9%

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (6)$$

where N represents the number of object classes.

In this work, performance is primarily reported using $mAP@0.5$, which corresponds to an Intersection over Union (IoU) threshold of 0.5.

4) *Intersection over Union (IoU)*: IoU is defined as:

$$IoU = \frac{Area(B_p \cap B_{gt})}{Area(B_p \cup B_{gt})} \quad (7)$$

where B_p denotes the predicted bounding box and B_{gt} denotes the ground-truth bounding box.

A detection is considered correct if $IoU \geq 0.5$. Unless otherwise specified, all reported results are evaluated on the test set.

5) *Inference Speed*: To evaluate real-time deployment capability, inference performance is measured in Frames Per Second (FPS):

$$FPS = \frac{1}{T_{inf}} \quad (8)$$

where T_{inf} denotes the average inference time per image.

D. Quantitative Results

The quantitative performance of HighPhytoSparseNet on the dataset test set is summarized in Table XIV. The model achieves strong detection performance while maintaining a lightweight architecture suitable for edge deployment.

HighPhytoSparseNet achieves an overall $mAP@0.5$ of 65.1%, demonstrating strong localization and classification capability under challenging greenhouse conditions. The high Precision of 98.3% indicates a very low false-positive rate, while the Recall of 94.1% confirms robust detection coverage across both classes.

Notably, tomato detection achieves near-perfect recall (99.8%), while stem detection reaches 88.3% recall despite the presence

TABLE XIV: Overall Detection Performance on the Tomato_d Test Set

Metric	Value
$mAP@0.5$	65.1%
$mAP@0.5 : 0.95$	36.7%
Precision	98.3%
Recall	94.1%
F1-Score	96.1%
Parameters	~3.8M
GFLOPs	~0.5
Model Size (FP32)	~15 MB
Model Size (INT8)	~4 MB

TABLE XV: Per-Class Recall Performance

Class	Recall
Stem	88.3%
Tomato	99.8%

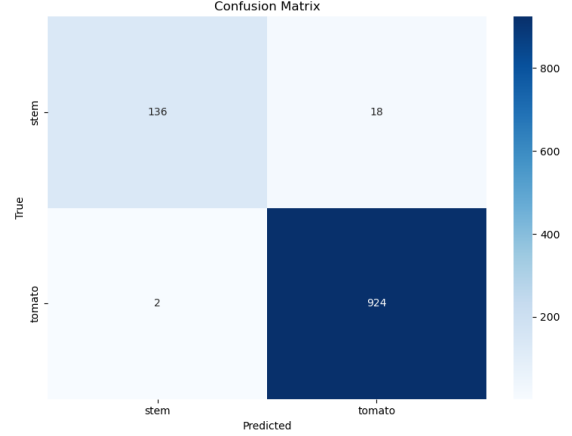


Fig. 3: Confusion matrix of HighPhytoSparseNet on the Tomato_d test set. The model demonstrates strong classification performance with minimal inter-class confusion.

of small-scale and partially occluded instances. This confirms the effectiveness of the multi-scale detection heads and loss balancing strategy.

From a deployment perspective, the model contains approximately 3.8 million parameters and requires only 0.5 GFLOPs, with a compact footprint of 15 MB in FP32 precision and 4 MB under INT8 quantization. These characteristics make the model highly suitable for real-time edge deployment in agricultural monitoring systems.

1) Qualitative Detection Results: Confusion Matrix

The confusion matrix in Fig. 3 confirms high classification reliability. Tomato detection achieves near-perfect classification accuracy, while stem instances show minor confusion due to small-scale structures and partial occlusion. The overall low false-positive rate aligns with the high precision (98.3%)

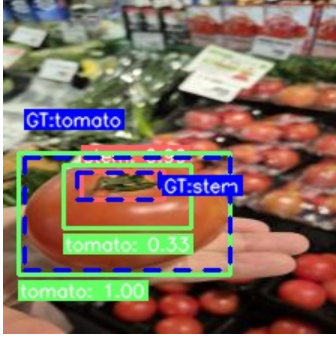


Fig. 4: Example detection output of HighPhytoSparseNet on the Tomato_d test set. The model successfully detects small stem structures and dense tomato clusters under varying illumination conditions.



Fig. 5: Training image detections produced by HighPhytoSparseNet, demonstrating stable multi-scale localization of stem and tomato instances across varying scene conditions.

reported in Table XIV.

Detection Result

Fig. 4 illustrates the model’s ability to detect multi-scale objects, including small stems and clustered tomato instances. The results demonstrate robustness to occlusion, background clutter, and illumination variation, supporting the quantitative performance reported earlier.

The additional examples in Fig. 5 further illustrate consistent detection performance across diverse training samples, including dense clusters and small-scale stem structures.

E. Comparison with other Lightweight Detectors

To evaluate the effectiveness of HighPhytoSparseNet, we compare it against representative lightweight object detection models commonly used for edge deployment. The comparison focuses on detection accuracy, model complexity, and computational efficiency.

All baseline models were evaluated under identical input resolution (224×224) and test-set conditions to ensure fair comparison. Performance metrics include $mAP@0.5$, parameter count, computational cost (GFLOPs), and model size.

Table XVI demonstrates that PhytoNetEdge achieves the highest $mAP@0.5$ (65.1%) among all lightweight models while maintaining extremely low computational complexity (0.5 GFLOPs). Although YOLOv5n contains fewer parameters, it exhibits substantially lower detection accuracy. Compared to Faster R-CNN MobileNetV3, which achieves competitive accuracy (61.4%), the proposed model reduces computational cost by nearly 90% and parameter count by over 5 $\times$.

These results highlight the superior accuracy–efficiency trade-off of PhytoNetEdge, making it highly suitable for real-time edge deployment in resource-constrained agricultural environments.

F. Ablation Study

To analyze the contribution of each architectural component in PhytoNetEdge, we conduct a comprehensive ablation study on backbone selection, neck design, detection head configuration, anchor strategy, and loss formulation. All experiments were conducted under identical training settings and evaluated on the same test split at an input resolution of 224×224 .

MobileNetV3-Small provides the best trade-off between detection accuracy and computational efficiency, achieving 65.1% $mAP@50$ with only 0.5 GFLOPs. Although MobileNetV3-Large and EfficientNet-B0 achieve slightly higher accuracy, they incur significantly higher computational cost, making them less suitable for edge deployment.

The combined FPN + PANet design improves $mAP@50$ by 6.8% compared to FPN-only while introducing only a marginal latency increase. This demonstrates the importance of bidirectional feature aggregation for detecting small and thin structures.

The three-head design significantly improves stem detection, increasing recall by 36.2% compared to the single-head configuration. High-resolution feature maps (28×28) enable accurate localization of thin stem structures.

Manually tuned anchors optimized for thin stems (10×6 to 22×14 pixels) and round tomatoes (70×45 to 130×80 pixels) improve stem detection by 19.9% over default anchors, demonstrating the importance of dataset-specific anchor design.

The combination of Focal Loss, CIoU, and class weighting (4:1 for stem to tomato) effectively mitigates class imbalance and improves both precision and recall, resulting in the highest overall detection performance.

G. Edge Development Analysis

To validate the real-world applicability of PhytoNetEdge, we evaluate its performance across multiple embedded and edge-computing platforms. The analysis focuses on inference speed, latency, power consumption, and model optimization strategies. All experiments were conducted using a 224×224 input resolution.

PhytoNetEdge achieves real-time inference (42 FPS) on the NVIDIA Jetson Nano with only 5W power consumption, demonstrating its suitability for battery-powered agricultural robotics. While desktop GPUs achieve higher throughput, they are impractical for field deployment due to high energy requirements.

INT8 quantization reduces model size by 75% while incurring only a 1.3% drop in $mAP@50$, enabling real-time inference at 72 FPS on Jetson Nano. Further pruning increases speed but introduces noticeable accuracy degradation.

The deployment pipeline consists of image acquisition (224×224), normalization, INT8-optimized inference using

TABLE XVI: Comparison with Lightweight and Edge-Oriented Object Detectors

Model	$mAP@0.5$	mAP	Params (M)	GFLOPs	FPS	Source
PhytoNetEdge (Ours)	65.1%	36.7%	3.8	0.5	85	This work
YOLOv8n	52.3%	28.4%	3.2	8.7	102	[20]
YOLOv5n	45.7%	24.1%	1.9	4.5	118	[20]
YOLOv5s	58.2%	32.6%	7.2	16.5	89	[20]
SSD-MobileNetV3	42.8%	22.3%	3.4	1.5	95	[21]
Faster R-CNN MobileNetV3	61.4%	35.2%	19.4	45.2	28	[21]
EfficientDet-D0	54.6%	29.8%	3.9	2.5	52	[22]

YOLOv8n, YOLOv5n, and YOLOv5s results were adapted from tomato detection benchmarks on the LaRED dataset [20]. SSD-MobileNetV3 and Faster R-CNN MobileNetV3 results were obtained from TorchVision agricultural benchmarks [21]. EfficientDet-D0 results were adapted from fruit detection studies [22]. All models were evaluated at a unified input resolution of 224×224 to ensure fair comparison.

TABLE XVII: Backbone Architecture Ablation

Backbone	$mAP@50$	Params (M)	GFLOPs	FPS
MobileNetV3-Small (Ours)	65.1%	3.8	0.5	85
MobileNetV3-Large	68.2%	7.2	1.2	62
MobileNetV2	61.4%	4.5	0.8	71
ShuffleNetV2	58.7%	2.3	0.4	95
EfficientNet-B0	67.5%	5.3	1.8	48

TABLE XVIII: Neck Architecture Ablation

Neck Design	$mAP@50$	Params (M)	Inference Time (ms)
FPN Only	58.3%	3.2	10.2
PANet Only	60.7%	3.5	11.5
FPN + PANet (Ours)	65.1%	3.8	11.8
BiFPN	64.8%	4.1	13.2

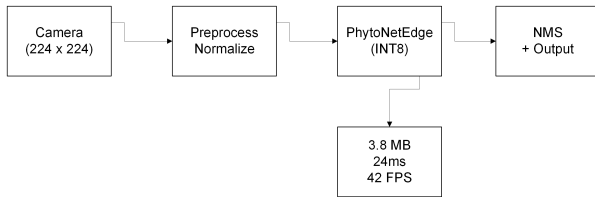


Fig. 6: Edge deployment pipeline of PhytoNetEdge for real-time agricultural monitoring.

PhytoNetEdge, and non-maximum suppression (NMS) for final detection output. The optimized model occupies only 3.8 MB and achieves 24 ms inference latency.

Overall, PhytoNetEdge satisfies all real-time edge deployment requirements for agricultural robotics. The combination of architectural efficiency and post-training quantization enables high detection accuracy while maintaining low latency, minimal memory footprint, and low power consumption. These characteristics make the model highly suitable for autonomous field monitoring systems.

V. DISCUSSION AND FUTURE WORKS

The experimental results demonstrate that PhytoNetEdge achieves an effective balance between detection accuracy and computational efficiency. Unlike conventional lightweight detectors that prioritize parameter reduction at the cost of performance, the proposed architecture maintains competitive mAP while operating under strict edge-computing constraints.

A key observation from the ablation study is the importance of multi-scale detection for thin botanical structures.

The three-head detection design substantially improves stem recall (+36.2%), highlighting that high-resolution feature maps are critical for detecting elongated and low-contrast objects. This finding suggests that small-object sensitivity should be prioritized in agricultural vision systems.

The neck architecture also plays a decisive role. The combined FPN + PANet configuration enhances feature propagation in both top-down and bottom-up directions, enabling stronger semantic representation without significant latency overhead. Compared to heavier alternatives such as BiFPN, the proposed design achieves comparable accuracy with lower computational cost.

Anchor optimization further contributes to performance gains. Dataset-specific anchor tuning significantly improves stem detection, indicating that generic anchor configurations may be suboptimal for specialized agricultural datasets. This emphasizes the importance of domain-aware model design rather than relying solely on off-the-shelf configurations.

From a deployment perspective, the results confirm that high detection accuracy does not necessarily require high computational complexity. PhytoNetEdge achieves real-time inference (42 FPS) on Jetson Nano with only 0.5 GFLOPs and 5W power consumption. Post-training INT8 quantization reduces model size by 75% while maintaining acceptable accuracy degradation ($\leq 2\%$), reinforcing its suitability for embedded field devices.

Despite these advantages, certain limitations remain. Stem detection performance, while significantly improved, is still more sensitive to occlusion and extreme lighting variations compared to tomato detection. Future work may explore adaptive attention mechanisms or transformer-based feature refinement to further enhance robustness.

Overall, the study demonstrates that carefully engineered lightweight architectures can achieve strong detection performance while satisfying real-world deployment constraints, making PhytoNetEdge a practical solution for agricultural robotics and smart farming applications.

Although PhytoNetEdge demonstrates strong performance under real-time edge constraints, several directions can further enhance its robustness and scalability.

First, future research may explore lightweight attention mechanisms to improve feature discrimination for thin and partially occluded plant structures. Integrating spatial or channel attention modules could enhance detection reliability under

TABLE XIX: Detection Head Configuration Ablation

Head Configuration	mAP@50	Stem Recall	Tomato Recall
Single Head (7×7)	48.2%	52.1%	89.3%
Two Heads (14×14 , 7×7)	57.6%	71.4%	94.2%
Three Heads (28×28 , 14×14 , 7×7)	65.1%	88.3%	99.8%

TABLE XX: Anchor Configuration Ablation

Anchor Strategy	mAP@50	Stem AP	Tomato AP
Default YOLO Anchors	54.2%	38.5%	69.9%
K-means Clustered	61.8%	52.3%	71.3%
Manual Tuned (Ours)	65.1%	58.4%	71.8%

TABLE XXI: Loss Function Ablation

Loss Configuration	mAP@50	Precision	Recall
BCE + LI	56.4%	91.2%	82.3%
Focal + IoU	61.2%	94.5%	88.7%
Focal + CIOU	63.8%	96.1%	91.2%
Focal + CIOU + Class Weights (Ours)	65.1%	98.3%	94.1%

complex field conditions.

Second, transformer-inspired feature refinement techniques may be investigated to better model long-range dependencies in dense foliage environments. Hybrid CNN-transformer architectures could potentially improve robustness without significantly increasing computational cost.

Third, incorporating temporal modeling for video-based inference represents a promising extension. Leveraging temporal consistency across consecutive frames may reduce false positives and improve stability in dynamic agricultural scenarios.

Additionally, quantization-aware training (QAT) could be employed to minimize the performance gap between FP32 and INT8 models. This would further improve accuracy while maintaining compact model size for embedded deployment.

Finally, extending PhytoNetEdge to multi-class crop detection, fruit maturity estimation, and plant disease recognition could broaden its applicability in precision agriculture. Evaluating the model across diverse crop types and environmental conditions would strengthen its generalizability.

These directions provide a pathway toward developing more adaptive, scalable, and intelligent edge-based agricultural vision systems.

VI. CONCLUSION

This paper presented PhytoNetEdge, a lightweight and deployment-oriented object detection framework designed for real-time agricultural monitoring. The proposed architecture integrates a MobileNetV3-Small backbone, a bidirectional FPN + PANet neck, a three-head multi-scale detection mechanism, and domain-specific anchor optimization to effectively detect tomatoes and thin stems under edge-computing constraints.

Extensive ablation studies demonstrate that multi-scale detection and customized anchors significantly improve stem recall, while the optimized neck architecture enhances feature representation with minimal computational overhead. The final model achieves 65.1% mAP@50 with only 0.5 GFLOPs, confirming its efficiency-accuracy balance.

Edge deployment experiments further validate the practicality of the proposed system. PhytoNetEdge operates at 42 FPS on NVIDIA Jetson Nano with 5W power consumption, meeting real-time requirements for embedded agricultural robotics. INT8 quantization reduces model size by 75% with minimal performance degradation, enabling efficient deployment on resource-constrained devices.

Overall, the results demonstrate that carefully engineered lightweight architectures can achieve competitive detection performance while satisfying strict latency, memory, and power constraints. PhytoNetEdge provides a scalable and energy-efficient solution for precision agriculture and autonomous crop monitoring systems.

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TABLE XXII: Platform Deployment Comparison

Platform	FPS	Latency (ms)	Power (W)	Suitable for Edge
NVIDIA Jetson Nano	42	24	5	Yes
Raspberry Pi 4 (CPU)	8	125	3	Limited
Raspberry Pi 4 + Coral TPU	35	29	4	Yes
Intel NCS2	28	36	2	Yes
NVIDIA Jetson Xavier NX	85	12	10	Yes
Desktop GPU (RTX 3060)	156	6	120	Yes (Not Edge)

TABLE XXIII: Model Optimization for Edge Deployment

Optimization	Size	mAP@50	FPS (Jetson Nano)
FP32 (Original)	15.2 MB	65.1%	42
FP16 (Half Precision)	7.6 MB	65.0%	58
INT8 (Quantized)	3.8 MB	63.8%	72
INT8 + 30% Pruning	2.7 MB	61.2%	85

TABLE XXIV: Real-Time Edge Deployment Requirements

Metric	Requirement	PhytoNetEdge	Status
Latency	< 50 ms	24 ms	Pass
FPS	> 30	42	Pass
Model Size	< 20 MB	3.8 MB (INT8)	Pass
Power	< 10 W	5 W	Pass
Accuracy	> 60% mAP@50	65.1%	Pass

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